

# TECHNICAL AND FINANCIAL OFFER

## Steel-framed sandwich-panel roof over the existing roof slab Administration Building, Regatta Tourist Village, Tripoli

|           |                                                       |              |                                                                           |
|-----------|-------------------------------------------------------|--------------|---------------------------------------------------------------------------|
| Offer No. | RTV-2026-01 Rev A (replaces the issue of 29 Sep 2026) | Date         | 29 September 2026                                                         |
| To        | Regatta Tourist Village - Administration              | Valid until  | 14 October 2026                                                           |
| From      | Eng. MALEK ABOZRAIG,<br>malek.abozraig@gmail.com      | Design basis | Design Rev 7 (reduced roof area per the architect's 2nd-level plan A_103) |

Dear Sirs,

Further to the survey of the existing administration building, the structural design completed for it and the architect's 2nd-level layout of 29 September 2026, which encloses a smaller area than first briefed, we are pleased to submit our revised offer for the supply, fabrication and erection of a steel roof structure with insulated sandwich-panel cladding over the existing roof slab, as described below and shown on the attached framing plan and 3D view. The offer is a lump sum covering all materials, workshop fabrication, transport, site works and erection listed in the scope, delivered as a complete and weather-tight roof.

### 1. The structure

The new roof is a light steel post-and-beam structure standing on the existing 300 mm roof slab, with one steel column on each of the 20 existing concrete columns under the enclosed area, so that all loads pass straight into the building's own structure, plus three light wind posts on the south walls. Level primary beams run along the column rows; sloping rafters span between them and carry cold-formed purlins and 50 mm insulated PIR sandwich panels on a single roof plane falling 6 % to the north, where two box gutters and four downpipes on the north facade take the rainwater away. Lateral stability is provided by six X-braced wall bays and a braced roof plane, designed for the Tripoli wind and seismic conditions. The stair well remains open with upstands and flashings; the south-west terrace and the terrace with the skylight in front of the east wing are not roofed, and the roof edges there sit on the existing column lines K19-K20 and K16-K17-K18. The roofed area is 316 m2; the clear height under the steel is 3.0 m at the low north eave and about 4.0 m at the south edges.

| Element              | Specification                                                                                                              |
|----------------------|----------------------------------------------------------------------------------------------------------------------------|
| Columns              | 20 no. HEA 140, S275, one over each existing concrete column under the roof, plus three wind posts                         |
| Primary beams        | IPE 300 (one 13.7 m eave beam IPE 330), S275, level, bolted cap plates on the columns                                      |
| Rafters and trimmers | IPE 240, S275, single roof plane at 6 % falling north, bolted fin plates                                                   |
| Purlins              | Z 200 x 2.0 cold-formed, S350GD galvanised, at 1.5 m centres (wall girts excluded)                                         |
| Bracing              | L 70 x 7 wall diagonals in 6 bays; M24 roof rods in 12 panels                                                              |
| Bases                | 4 post-installed dia 16 rebars per column into the existing column head, grouted shear keys, 20 mm plates, 25 mm grout bed |
| Roof cladding        | PIR sandwich panels 50 mm, colour-coated steel both faces, with ridge, eave and verge flashings and well upstands          |
| Rainwater            | Two box gutters 150 x 100 with overflows, four 100 mm downpipes on the north facade                                        |
| Steel weight         | about 15 tonnes, all steel hot-dip galvanised or shop-primed and painted as agreed                                         |

## 2. Scope of supply and works

- Detailed fabrication drawings and the bill of materials based on the approved design (Rev 7, reduced roof area).
- Supply of all structural steel sections, plates, bolts, rebars, anchors and cold-formed purlins and girts.
- Workshop fabrication, surface protection, transport to site and offloading.
- Site survey of the slab top, scanning of the 20 column heads, drilling and injection of the anchors, coring and grouting of the shear keys, and proof tests on three bars.
- Erection of columns, beams, rafters, bracing and purlins with all connections, including the temporary bracing needed during erection.
- Supply and fixing of the roof sandwich panels with all flashings, well upstands, sealants and fasteners.
- Supply and fixing of gutters, downpipes and the associated brackets and flashings.
- Site management, safety measures, cleaning of the site and handover with as-built drawings.

## 3. Not included

- Wall cladding of any kind and its girts or fixings (the architect's drawings show 20 cm gypsum walls; to be offered separately once the wall system is selected).
- Verification of the existing building (as-built drawings, slab-top survey drawing, adequacy statement of the existing columns and foundations) and any strengthening it may require.
- Building permits, authority fees, independent checking and any works to the existing slab or facade beyond the anchors, keys and downpipe fixings.
- Electrical, mechanical, lighting, ceilings, floors and finishes under the new roof.

## 4. Programme

The works are planned over about three weeks from receipt of the advance payment, subject to site access and to the scan results at the column heads.

| Week | Activity                                                                                                                   |
|------|----------------------------------------------------------------------------------------------------------------------------|
| 1    | Fabrication drawings approved; slab survey and column-head scans; steel and panel procurement; workshop fabrication starts |
| 2    | Fabrication completed and delivered; anchors drilled and injected, shear keys cored and grouted; columns and beams erected |
| 3    | Rafters, bracing and purlins completed; roof panels, flashings, gutters and downpipes fixed; cleaning and handover         |

## 5. Price

|                                                                                                                                                                                           |                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>Lump-sum price for the complete scope in section 2</b>                                                                                                                                 | <b>LYD 284,000</b> |
| Libyan Dinars, fixed for the validity period, inclusive of materials, fabrication, transport, site works, erection and supervision. Taxes and fees, if any, are for the client's account. |                    |

## 6. Payment terms

| Stage                                              | Share | Amount, LYD |
|----------------------------------------------------|-------|-------------|
| Advance on signing the order; starts the programme | 50 %  | 142,000     |
| On start of erection on site                       | 25 %  | 71,000      |
| On completion and handover                         | 25 %  | 71,000      |

## 7. Conditions

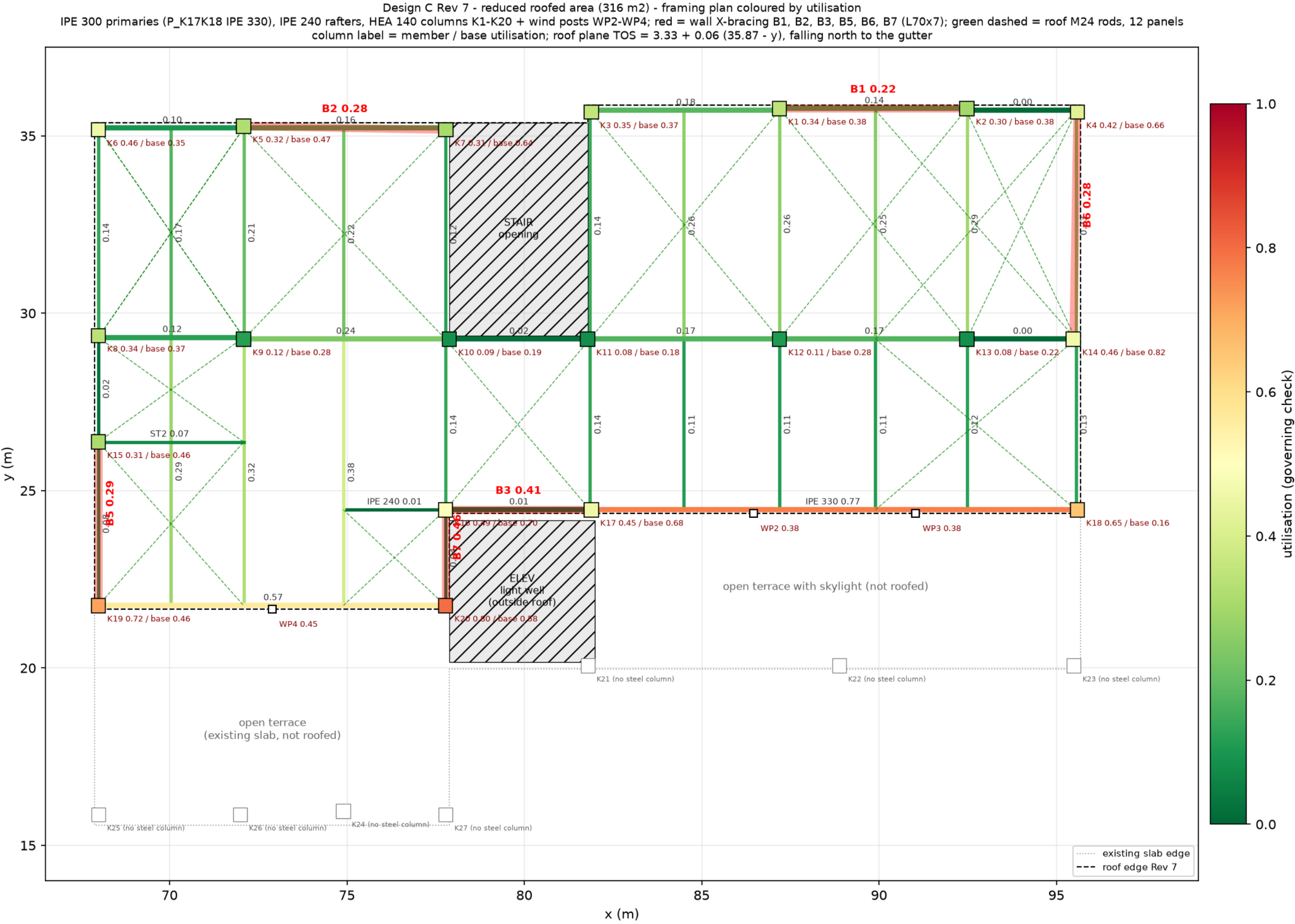
- Validity: this offer is valid for 15 days from its date, until 14 October 2026, because of the movement of steel prices.
- Warranty: 12 months from handover on the steel structure, panel fixings and workmanship; panel and coating warranties as per the manufacturer.
- The design is based on the assumptions listed in the design report (Rev 7) and the load basis Rev 3. The client provides as-built information on the existing building and free access to the roof and to the north facade for the downpipes; anything found at the column heads that differs from the assumptions (bar positions, concrete quality, slab thickness) is dealt with by the fallback base details of the design set and, if it changes the scope, by a written variation before work continues.
- The client confirms that the six braced wall bays shown on the framing plan (K1-K2, K5-K7, K16-K17, K15-K19, K4-K14, K16-K20) remain free of doors and windows, and that the architect adjusts the 2nd-level wall lines to the roof edges shown.
- Materials: structural steel S275 to EN 10025, bolts 8.8, cold-formed purlins S350GD, PIR panels with a manufacturer's certificate; anchor system with a European Technical Assessment for post-installed rebar.
- Design standards: Eurocodes EN 1990, 1991, 1993 and 1998 with parameters for Tripoli, as recorded in the design report.
- Any change to the roof plane, the openings, the cladding type or the scope after the order is priced separately.
- The offer excludes anything not expressly stated in section 2.

## 8. Attachments

- Attachment A: roof framing plan, design Rev 7 (A3).
- Attachment B: 3D view of the steel structure, design Rev 7.
- The detailed drawing set (fabrication and erection drawings, A2) is produced after the order on the approved Rev 7 design; the Rev 6a set RTV-ST-S00 to S06 remains available as reference for the typical details.

We remain at your disposal for any clarification and look forward to working with you on this project.

| For the offer                                                  | For the client (acceptance)                   |
|----------------------------------------------------------------|-----------------------------------------------|
| Eng. MALEK ABOZRAIG<br>malek.abozraig@gmail.com<br>Date: ..... | Name: .....<br>Position: .....<br>Date: ..... |



Steel roof, design Rev 7 - reduced roofed area (316 m<sup>2</sup>): view from the south-west, roof panels hidden

